Software Requirements Specification

for

AXIS AI

Agentic Multimodal Framework for Intelligent Office Workflow Automation

Version 1.0

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Revision History

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Chapter 1

Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define the functional and non-functional requirements of AXIS (Agentic eXecution Intelligence Suite), an AI-powered framework for intelligent office workflow automation. It describes the system's objectives, core features, constraints, and technical requirements to guide its design, development, testing, and evaluation. AXIS aims to provide a unified workspace that automates tasks across Microsoft Word, Excel, and PowerPoint using natural language and voice commands, leveraging AI agents, Large Language Models (LLMs), multimodal document understanding, and OCR. This document serves as a common reference for developers, testers, and stakeholders to ensure that the system meets its intended objectives while addressing reliability, security, explainability, and human oversight.

1.2 Document Conventions

This document follows a structured Software Requirements Specification format inspired by the IEEE SRS structure.

- Section numbering is used for easy reference.
- Mandatory functional requirements use the phrase "The system shall".
- Requirements have unique identifiers such as AX-REQ-01.
- "Should" indicates a desirable capability that depends on project scope and resources.
- AI-generated results may contain errors and should be reviewed before important use.

1.3 Intended Audience and Reading Suggestions

This document is intended for project guides, review committee members, developers, students, and testers involved in the design, development, and evaluation of AXIS. Readers are advised to begin with the Introduction and Overall Description to understand the project's objectives, scope, and general functionality, followed by the System Features section for detailed functional requirements. The remaining sections describe the external interfaces, non-functional requirements, constraints, and other technical details necessary for implementing and validating the system.

1.4 Product Scope

AXIS (Agentic eXecution Intelligence Suite) is an AI-powered office workflow automation platform designed to simplify and automate repetitive tasks across Microsoft Word, Excel, and PowerPoint. It uses Large Language Models (LLMs), agentic AI, multimodal document understanding, and Optical Character Recognition (OCR) to support document generation and summarization, spreadsheet analysis, automated dashboard creation, presentation generation,

and information extraction from scanned documents. The system also incorporates voice and natural language commands, multi-agent collaboration, retrieval-augmented generation (RAG), multilingual assistance, personalized workflows, anomaly detection, document comparison, data privacy mechanisms, explainable AI, and human-in-the-loop approval. By integrating these capabilities into a unified workspace, AXIS aims to reduce manual effort, improve productivity, and support efficient decision-making while maintaining user oversight of important AI-generated outputs and actions.

1.5 References

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Chapter 2

Overall Description

2.1 Product Perspective

AXIS (Agentic eXecution Intelligence Suite) is a proposed AI-powered office workflow automation platform designed to provide a unified environment for performing tasks across Microsoft Word, Excel, and PowerPoint. Unlike conventional office applications that primarily require users to perform tasks manually or rely on separate AI assistants for individual operations, AXIS aims to coordinate multiple AI agents to execute complex workflows through natural language and voice commands. The system integrates Large Language Models (LLMs), agentic AI, multimodal document understanding, Optical Character Recognition (OCR), and intelligent data processing to support document generation and summarization, spreadsheet analysis, dashboard creation, presentation generation, and cross-application automation. It is envisioned as a web-based application with a user interface, backend services, database, and AI processing components that work together to interpret user requests, manage workflows, and generate relevant outputs. Additional capabilities, including Retrieval-Augmented Generation (RAG), multilingual assistance, anomaly detection, document comparison, explainable AI, data privacy, and human-in-the-loop approval, are intended to improve usability, reliability, and control. AXIS is designed to complement existing office applications by integrating their capabilities within a centralized, intelligent workflow environment.

2.2 Product Functions

The main functions include:

- **Voice Recognition and Speech Commands:** Enables users to interact with the system through voice commands to initiate office tasks.
- **Agentic AI Workflow Automation:** Interprets user requests, plans task sequences, and automates office workflows using AI agents.
- **Multi-Agent Collaboration:** Coordinates specialized AI agents to perform different tasks and collaborate towards completing complex workflows.
- **AI Document Generation and Summarization:** Generates, edits, and summarizes documents based on user instructions and provided information.
- **Intelligent Spreadsheet Analysis:** Processes spreadsheet data, identifies patterns, and assists users in analyzing structured information.
- **Automatic Dashboard and Visualization Generation:** Converts analyzed data into suitable charts, visualizations, and dashboards.
- **OCR and Handwritten Document Processing:** Extracts text and relevant information from scanned, photographed, and supported handwritten documents.
- **Multimodal Document Understanding:** Analyzes document content, including text, layouts, tables, and visual elements, to support information extraction and interpretation.
- **Automatic Presentation Generation:** Creates structured presentations with relevant content and visual elements based on user requirements.

- **Cross-Application Workflow Automation and Explainable AI:** Coordinates tasks across office applications and provides explanations of AI-generated decisions and actions.
- **Retrieval-Augmented Generation (RAG) for Office Knowledge:** Retrieves relevant information from available documents and knowledge sources to provide context-aware responses.
- **Personalized AI and Workflow Learning:** Adapts assistance to user preferences and workflow requirements based on available preferences and feedback.
- **Multilingual Office AI:** Supports office-related interactions and content generation in supported languages.
- **AI-Based Data Privacy and Security:** Identifies sensitive information and supports appropriate data-protection measures during processing.
- **Intelligent Anomaly and Error Detection:** Identifies unusual data patterns, inconsistencies, and potential errors in office data.
- **Human-in-the-Loop AI and Approval:** Allows users to review, modify, or approve selected AI-generated outputs and actions before execution.
- **AI Formula and Code Generation:** Generates spreadsheet formulas, queries, or code from natural-language instructions.
- **Intelligent Document Comparison and Change Detection:** Compares documents to identify textual similarities, differences, and modifications between versions.

2.3 User Classes and Characteristics

The AXIS (Agentic eXecution & Intelligence Suite) system is intended for users with different levels of technical knowledge and office productivity requirements. The primary user classes are:

- **Students and Researchers:** Use AXIS to prepare reports, summarize documents, analyze datasets, generate visualizations, and create presentations, reducing the time required for academic and research-related tasks.
- **Educators and Faculty Members:** Use the system to prepare educational materials, organize information, analyze academic data, and generate presentations and reports.
- **Office Professionals and Employees:** Use AXIS to automate routine documentation, spreadsheet processing, data analysis, and presentation preparation through natural language or voice commands.
- **Business Analysts and Managers:** Use the system to analyze business data, identify anomalies, generate dashboards, and obtain relevant insights to support decision-making.
- **Administrators and Organizational Users:** Use AXIS to process organizational documents, retrieve relevant information, and automate repetitive administrative workflows while considering data privacy and security requirements.
- **System Administrators and Developers:** Maintain system configuration, manage integrations, monitor system performance, and troubleshoot technical issues. Developers may also extend the platform by integrating additional AI models, tools, and workflow capabilities.

The system is designed to be accessible to both technical and non-technical users. Users can interact with AXIS through natural language or voice commands without requiring advanced programming knowledge. Depending on their roles and permissions, users may review

AI-generated outputs, approve selected actions, and provide feedback to improve the relevance and effectiveness of the system.

2.4 Operating Environment

- **Client Environment:** A desktop or laptop computer with a modern web browser such as Google Chrome, Microsoft Edge, or Mozilla Firefox.
- **Frontend:** React.js for developing the user interface and enabling interaction through natural language and voice commands.
- **Backend:** Python with FastAPI for handling user requests, coordinating AI agents, managing workflows, and communicating with external AI services.
- **Database:** MongoDB for storing user information, application data, preferences, and relevant workflow details.
- **AI Processing Environment:** Compatible Large Language Models (LLMs), agentic AI frameworks, and supporting tools for document generation, spreadsheet analysis, OCR, multimodal processing, and workflow automation.
- **Supporting Libraries and Services:** Libraries such as PaddleOCR, Pandas, Plotly, and OpenPyXL may be used for text extraction, data processing, visualization, and spreadsheet operations, depending on the implemented functionality.
- **Network Requirements:** An active internet connection is required for accessing the web application and any cloud-hosted AI models or external services used by the system.

The system is designed to operate independently of a specific desktop operating system, provided that a supported web browser and a stable internet connection are available. Actual resource requirements and compatibility may vary depending on the AI models, integrations, and features enabled.

2.5 Design and Implementation Constraints

- **Technology Stack:** The system is intended to use React.js for the frontend, Python with FastAPI for backend services, and MongoDB for data storage. The implementation must remain compatible with the selected technologies and their dependencies.
- **AI Model Dependency:** AI-based features depend on the capabilities, availability, and limitations of the selected Large Language Models (LLMs) and supporting AI services. Model-generated outputs may require user verification.
- **External Service Dependency:** Features that rely on cloud-based AI models, speech recognition, or third-party integrations may require an active internet connection and valid API credentials. Service availability and usage limits may affect system performance.
- **Document Compatibility:** Document processing and generation must consider supported file formats, file sizes, document structures, and the capabilities of the selected processing libraries.
- **Data Privacy and Security:** User documents and sensitive information must be handled securely. Appropriate access controls and data-protection measures must be considered when processing or transmitting information to external services.
- **Processing and Resource Limitations:** Complex workflows involving large documents, spreadsheets, OCR, or multiple AI agents may require significant processing time and computational resources.

- **Integration Constraints:** Cross-application automation depends on the availability and compatibility of supported APIs, libraries, and office application interfaces.
- **Human Oversight:** Critical operations and selected AI-generated outputs must support user review and approval before execution, wherever required by the workflow.

2.6 User Documentation

The system should provide a user guide, supported-file information, examples of text and voice commands, instructions for reviewing and downloading outputs, and troubleshooting messages for failed tasks.

2.7 Assumptions and Dependencies

The system assumes users submit files they are authorized to process and provide sufficiently clear instructions. Operation depends on the configured language model, speech-recognition component, OCR engine, file-processing libraries, backend, and network. Multilingual, handwriting, and multimodal performance depends on selected models. The first prototype may prioritize a subset of features based on time and resources.

Chapter 3

External Interface Requirements

3.1 User Interfaces

The system is expected to provide:

- **Login and Registration Interface:** Allows users to register, log in, and access the system securely.
- **Dashboard:** Provides a centralized workspace for accessing available features, viewing recent activities, and initiating office automation tasks.
- **Natural Language Input Interface:** Allows users to enter instructions or prompts describing the tasks they want the system to perform.
- **Voice Command Interface:** Enables users to provide spoken instructions for supported office automation operations.
- **File Upload Interface:** Allows users to upload supported documents, spreadsheets, presentations, scanned files, and images for processing.
- **AI Output and Preview Interface:** Displays generated documents, summaries, spreadsheet analysis, charts, presentations, and other AI-generated results for user review.
- **Workflow Monitoring Interface:** Displays the progress and status of ongoing automation tasks and provides relevant feedback when a task is completed or encounters an error.
- **Review and Approval Interface:** Allows users to inspect, modify, approve, or reject selected AI-generated outputs and proposed actions before execution.
- **Settings and Preferences Interface:** Enables users to manage supported application preferences, language options, and personalization settings.

3.2 Hardware Interfaces

- **Computer System:** A desktop or laptop computer with sufficient processing capacity and memory to run a modern web browser and access the application.
- **Microphone:** A built-in or external microphone shall be required for voice-based interaction, where supported.
- **Speakers or Headphones:** Audio output devices may be used for receiving spoken responses or audio feedback, if implemented.
- **Camera or Scanner:** A camera or scanner may be used to capture physical documents for OCR and document-processing tasks. Alternatively, users may upload existing digital files.
- **Network Interface:** A network adapter and internet connectivity shall be required to access the web application and communicate with cloud-hosted AI models or external services.

3.3 Software Interfaces

- **Operating System:** The application shall be accessible through a supported web browser on commonly used operating systems, including Windows, Linux, and macOS.
- **Frontend Technologies:** React.js shall be used to develop the user interface and support user interaction with the system.
- **Backend Technologies:** Python and FastAPI shall handle user requests, application logic, AI processing coordination, and communication between system components.
- **Database Management System:** MongoDB shall be used to store user information, application data, preferences, and relevant workflow details.
- **Large Language Models (LLMs):** Compatible AI model APIs or locally hosted models shall support natural language understanding, content generation, summarization, and other AI-assisted tasks, depending on the selected implementation.
- **AI Agent Frameworks:** Agent orchestration frameworks, such as LangGraph or CrewAI, may be used to coordinate specialized agents and manage multi-step workflows.
- **OCR and Document Processing:** PaddleOCR and other compatible libraries may be used to extract text from scanned documents and images.
- **Data Analysis and Visualization:** Pandas, OpenPyXL, and Plotly may be used for spreadsheet processing, data analysis, Excel file operations, and chart generation.
- **Office Application Interfaces:** Supported APIs, libraries, or integration mechanisms shall be used where necessary to process or generate compatible Word documents, Excel spreadsheets, and PowerPoint presentations.
- **External Service Interfaces:** The system may communicate with third-party AI services and other external APIs through appropriate interfaces, subject to service availability, authentication, and usage restrictions.

3.4 Communications Interfaces

- Deployed client-server communication should use HTTPS.
- External API requests shall use the provider's required authentication.
- Long-running tasks should display status or progress where possible.
- The system shall display understandable errors when requests fail.
- API credentials shall not be exposed in frontend code.

Chapter 4

System Features

4.1 Voice Recognition and Speech Commands

4.1.1 Description and Priority

Allows users to give spoken commands. A speech-recognition component converts audio to text, which is then passed to the intent-understanding workflow. Priority: High.

4.1.2 Stimulus/Response Sequences

1. The user activates voice input and speaks a command.
2. The system captures audio and sends it to the configured speech-recognition component.
3. The system receives and displays the transcription where supported.
4. The transcription is passed to the command-understanding component.
5. The system informs the user if capture or transcription fails.

4.1.3 Functional Requirements

- **AX-REQ-01:** The system shall accept voice input when a supported microphone and speech-recognition component are available.
- **AX-REQ-02:** The system shall convert captured speech into text using the configured component.
- **AX-REQ-03:** The system shall notify the user when audio cannot be captured or transcribed.
- **AX-REQ-04:** The system should allow the user to review or correct transcription before consequential execution.

4.2 Agentic AI Workflow Automation

4.2.1 Description and Priority

Coordinates multi-step office tasks from a single request by decomposing the request, selecting configured tools, tracking execution, and combining results. Priority: High.

4.2.2 Stimulus/Response Sequences

1. The user submits a multi-step request.
2. The system identifies the outcome and breaks the request into tasks.
3. The orchestrator selects available agents or tools.
4. The system executes steps and tracks their status.
5. The system returns outputs and identifies failed or incomplete steps.

4.2.3 Functional Requirements

- **AX-REQ-05:** The system shall decompose supported multi-step requests into tasks.
- **AX-REQ-06:** The orchestrator shall invoke only configured and authorized tools.
- **AX-REQ-07:** The system shall track workflow-step status.
- **AX-REQ-08:** The system shall not report a failed operation as completed.

4.3 Multi-Agent Collaboration

4.3.1 Description and Priority

Enables specialist agents or components to handle tasks such as document processing, spreadsheet analysis, presentation generation, and output review. Priority: High.

4.3.2 Stimulus/Response Sequences

1. The orchestrator identifies tasks required by a request.
2. It assigns tasks to configured agents or tools.
3. Each component returns a result or error.
4. The orchestrator passes relevant results to dependent tasks.
5. The system combines results and reports unresolved failures.

4.3.3 Functional Requirements

- **AX-REQ-09:** The system should support specialized task roles for implemented office operations.
- **AX-REQ-10:** The orchestrator shall pass required inputs and context to assigned tasks.
- **AX-REQ-11:** The system shall associate each task with its result or error status.
- **AX-REQ-12:** The system shall identify agent failures.

4.4 AI Document Generation and Summarization

4.4.1 Description and Priority

Generates, summarizes, rewrites, and structures document content from user instructions or source material. Priority: High.

4.4.2 Stimulus/Response Sequences

1. The user enters a document request or uploads source material.
2. The system identifies the requested document type and extracts available content.
3. The language model generates or transforms content.
4. The system displays a draft and creates a downloadable document where supported.
5. The user reviews the result.

4.4.3 Functional Requirements

- **AX-REQ-13:** The system shall generate document content from a supported natural-language request.
- **AX-REQ-14:** The system shall summarize supported source content.
- **AX-REQ-15:** The system should support rewriting and structuring content.
- **AX-REQ-16:** The system shall allow review of generated content before important external use.

4.5 Intelligent Spreadsheet Analysis

4.5.1 Description and Priority

Reads supported spreadsheet data and provides summaries, calculations, trends, comparisons, and answers to user questions. Priority: High.

4.5.2 Stimulus/Response Sequences

1. The user uploads a spreadsheet and submits an analysis request.
2. The system reads available sheets, columns, and values.
3. The analysis component performs relevant calculations or summaries.
4. The system presents results and a natural-language explanation.
5. The system identifies missing or invalid data when it affects the analysis.

4.5.3 Functional Requirements

- **AX-REQ-17:** The system shall read supported spreadsheet formats.
- **AX-REQ-18:** The system shall provide analysis based on available data.
- **AX-REQ-19:** The system should identify missing or ambiguous data affecting a result.
- **AX-REQ-20:** The system should distinguish computed values from generated explanations.

4.6 Automatic Dashboard and Visualization Generation

4.6.1 Description and Priority

Creates charts, KPIs, and dashboard-style summaries from structured data. Priority: High.

4.6.2 Stimulus/Response Sequences

1. The user requests a chart or dashboard.
2. The system identifies relevant data fields and a suitable visualization.
3. The visualization component creates the chart or summary view.
4. The system displays the result with labels and explanation where supported.

4.6.3 Functional Requirements

- **AX-REQ-21:** The system shall generate supported charts from selected structured data.
- **AX-REQ-22:** The system shall label visualizations using available field names and units.
- **AX-REQ-23:** The system should provide a short explanation of the visualization.
- **AX-REQ-24:** The system shall inform the user when data is insufficient.

4.7 OCR and Handwritten Document Processing

4.7.1 Description and Priority

Extracts text from supported scanned documents and images. Handwriting recognition depends on the selected OCR engine and input quality. Priority: High.

4.7.2 Stimulus/Response Sequences

1. The user uploads a supported scan or image.
2. The system validates the file and sends it to the OCR component.
3. The OCR component returns extracted text.
4. The system displays the text or passes it to a requested task.
5. The system reports extraction failures where detectable.

4.7.3 Functional Requirements

- **AX-REQ-25:** The system shall extract text from supported images and scanned documents.
- **AX-REQ-26:** The system should preserve text order and page structure where supported.
- **AX-REQ-27:** The system shall inform the user when extraction fails.
- **AX-REQ-28:** The system shall not represent uncertain OCR output as guaranteed accurate.

4.8 Multimodal Document Understanding

4.8.1 Description and Priority

Processes supported combinations of text, images, tables, charts, and document pages. Capabilities depend on the configured model and preprocessing tools. Priority: High.

4.8.2 Stimulus/Response Sequences

1. The user uploads a supported document or image and asks a question.
2. The system extracts or processes available textual and visual information.
3. The AI component generates a response using the available context.
4. The system presents the answer and indicates unreadable content where relevant.

4.8.3 Functional Requirements

- **AX-REQ-29:** The system shall process multimodal input types supported by configured components.
- **AX-REQ-30:** The system should answer questions about accessible document content.
- **AX-REQ-31:** The system shall inform the user when content is unsupported or unreadable.

4.9 Automatic Presentation Generation

4.9.1 Description and Priority

Generates presentation outlines, slide titles, slide content, and supported presentation files from a prompt, document, or analysis result. Priority: High.

4.9.2 Stimulus/Response Sequences

1. The user specifies a topic or supplies a source document and presentation requirements.
2. The system creates an outline and organizes content into slides.
3. The presentation component applies supported layouts and visual elements.
4. The system creates a preview or downloadable file where supported.
5. The user reviews the presentation.

4.9.3 Functional Requirements

- **AX-REQ-32:** The system shall generate a slide outline from a supported topic or source document.
- **AX-REQ-33:** The system shall generate slide titles and content based on the outline.
- **AX-REQ-34:** The system should support PPTX export if the implementation provides it.
- **AX-REQ-35:** Speaker notes or visual assets may be supported if implemented.

4.10 Cross-Application Automation and Explainable AI

4.10.1 Description and Priority

Connects tasks across document, spreadsheet, and presentation workflows. It passes intermediate results between tasks and explains key results where supported. Priority: High.

4.10.2 Stimulus/Response Sequences

1. The user requests a workflow involving two or more office operations.
2. The system processes the source file and produces intermediate results.
3. The workflow passes relevant results to subsequent tasks.
4. The system returns outputs and summarizes steps performed.
5. Where supported, the system explains important calculations, data sources, or decisions.

4.10.3 Functional Requirements

- **AX-REQ-36:** The system shall pass relevant intermediate results between configured office tasks.
- **AX-REQ-37:** The system shall support a workflow combining at least two implemented office operations.
- **AX-REQ-38:** The system should explain key results or processing steps where available.
- **AX-REQ-39:** The system shall not imply that an explanation guarantees correctness.

4.11 Retrieval-Augmented Generation (RAG) for Office Knowledge

4.11.1 Description and Priority

Retrieval-Augmented Generation (RAG) retrieves relevant information from available documents and knowledge sources and supplies it as context to an AI model. This helps the system generate context-aware responses to user queries about office documents and organizational information. Priority: High.

4.11.2 Stimulus/Response Sequences

1. The user submits a question related to available documents or knowledge sources.
2. The system searches the configured knowledge repository for relevant information.
3. The retrieval component selects relevant content and provides it to the AI model.
4. The AI model generates a response using the retrieved context.
5. The system displays the response and identifies supporting sources where available.

4.11.3 Functional Requirements

- **AX-REQ-40:** The system shall retrieve relevant information from configured knowledge sources for supported queries.
- **AX-REQ-41:** The system shall provide retrieved context to the AI model when generating context-aware responses.
- **AX-REQ-42:** The system should identify supporting documents or source references where available.
- **AX-REQ-43:** The system shall indicate when sufficient relevant information cannot be retrieved to answer a query reliably.

4.12 Personalized AI and Workflow Learning

4.12.1 Description and Priority

Personalized AI and Workflow Learning adapts system assistance according to user preferences, previous interactions, and feedback where such information is available. The feature aims to improve the relevance of generated content and streamline frequently performed office tasks. Priority: Medium.

4.12.2 Stimulus/Response Sequences

1. The user configures preferences or provides feedback on an AI-generated result.
2. The system records supported preferences and relevant feedback.
3. The system uses available preference information when processing subsequent requests.
4. The system generates outputs or suggests workflows adapted to the user's preferences.
5. The user reviews the result and may update the preferences.

4.12.3 Functional Requirements

- **AX-REQ-44:** The system shall allow users to configure supported personalization preferences.
- **AX-REQ-45:** The system should use available user preferences to adapt supported outputs and workflows.
- **AX-REQ-46:** The system should allow users to update or reset their saved preferences.
- **AX-REQ-47:** The system shall not claim to have learned a preference or workflow unless the relevant information has been stored or processed.

4.13 Multilingual Office AI

4.13.1 Description and Priority

Multilingual Office AI enables users to interact with AXIS and generate office-related content in supported languages. Depending on the configured language models or translation services, the feature can assist with multilingual document creation, summarization, and information processing. Priority: Medium.

4.13.2 Stimulus/Response Sequences

1. The user submits a request or document in a supported language.
2. The system identifies or receives the requested language.
3. The language-processing component processes the input.
4. The system generates a response or requested office content in the selected language.
5. The system informs the user if the requested language or operation is unsupported.

4.13.3 Functional Requirements

- **AX-REQ-48:** The system shall process requests in languages supported by its configured language-processing components.
- **AX-REQ-49:** The system should generate or transform office content into a user-selected supported language.
- **AX-REQ-50:** The system should preserve the original meaning of content during supported translation or rewriting operations.
- **AX-REQ-51:** The system shall notify the user when a requested language operation is unsupported or cannot be completed.

4.14 AI-Based Data Privacy and Security

4.14.1 Description and Priority

AI-Based Data Privacy and Security supports the identification and protection of sensitive information in documents and other supported data. The feature may include sensitive-data detection, masking, and access-control mechanisms to reduce the risk of unauthorized disclosure during processing. Priority: High.

4.14.2 Stimulus/Response Sequences

1. The user uploads a document or initiates a data-processing task.
2. The system applies the configured security checks and sensitive-information detection mechanisms.
3. The system identifies information that may require protection.
4. Where supported, the system masks sensitive information or requests user review.
5. The system processes the permitted content and reports relevant warnings or errors.

4.14.3 Functional Requirements

- **AX-REQ-52:** The system shall apply configured authentication and authorization checks to protected operations.
- **AX-REQ-53:** The system should identify supported categories of sensitive information in documents or data.
- **AX-REQ-54:** The system should support masking or redaction of detected sensitive information where implemented.
- **AX-REQ-55:** The system shall not claim that all sensitive information has been detected or removed when detection or masking may be incomplete.

4.15 Intelligent Anomaly and Error Detection

4.15.1 Description and Priority

Intelligent Anomaly and Error Detection identifies unusual patterns, inconsistent values, missing data, and potential errors in supported office data, particularly spreadsheets. The feature assists users in reviewing data quality and identifying issues that may affect subsequent analysis or decision-making. Priority: High.

4.15.2 Stimulus/Response Sequences

1. The user uploads a supported spreadsheet or requests a data-quality check.
2. The system reads the available data and identifies applicable validation rules.
3. The detection component examines the data for unusual patterns and inconsistencies.
4. The system displays potential anomalies with relevant explanations where available.
5. The user reviews the findings and decides whether corrective action is required.

4.15.3 Functional Requirements

- **AX-REQ-56:** The system shall examine supported data for configured validation errors and inconsistencies.
- **AX-REQ-57:** The system should identify unusual data patterns using implemented detection methods.
- **AX-REQ-58:** The system shall present detected issues with relevant values or locations where available.
- **AX-REQ-59:** The system shall distinguish potential anomalies from confirmed errors and shall not modify source data without authorization.

4.16 Human-in-the-Loop AI and Approval

4.16.1 Description and Priority

Human-in-the-Loop AI and Approval allows users to review, modify, approve, or reject selected AI-generated outputs and proposed actions before execution. This feature provides human oversight for workflows that require confirmation and helps prevent unintended changes or operations. Priority: High.

4.16.2 Stimulus/Response Sequences

1. The system generates an output or proposes an action requiring approval.
2. The system presents the proposed output or action to the user.
3. The user reviews the information and may modify, approve, or reject it.
4. The system proceeds with the approved action or stops the rejected operation.
5. The system displays the resulting status to the user.

4.16.3 Functional Requirements

- **AX-REQ-60:** The system shall present selected outputs or proposed actions for user approval when the workflow requires it.
- **AX-REQ-61:** The system shall allow authorized users to approve or reject pending actions.
- **AX-REQ-62:** The system should allow users to modify reviewable outputs before approval where supported.
- **AX-REQ-63:** The system shall not execute an action requiring approval until the required approval has been received.

4.17 AI Formula and Code Generation

4.17.1 Description and Priority

AI Formula and Code Generation converts natural-language instructions into relevant spreadsheet formulas, queries, or supported code. The feature assists users in performing calculations and automating data-processing tasks without requiring them to write every formula or code segment manually. Priority: High.

4.17.2 Stimulus/Response Sequences

1. The user describes a calculation, data-processing task, or desired spreadsheet result.
2. The system identifies the required operation and available data context.
3. The AI component generates a suitable formula or supported code.
4. The system displays the generated formula or code and an explanation where available.
5. The user reviews the result and tests or applies it as appropriate.

4.17.3 Functional Requirements

- **AX-REQ-64:** The system shall generate supported spreadsheet formulas or code from natural-language instructions.
- **AX-REQ-65:** The system should use available information about spreadsheet structure and data when generating formulas.
- **AX-REQ-66:** The system should provide an explanation of the generated formula or code where supported.
- **AX-REQ-67:** The system shall allow users to review generated formulas or code before consequential execution or application.

4.18 Intelligent Document Comparison and Change Detection

4.18.1 Description and Priority

Intelligent Document Comparison and Change Detection compares supported document versions to identify additions, deletions, modifications, and other relevant differences. The feature helps users review revisions and understand how document content has changed between versions. Priority: Medium.

4.18.2 Stimulus/Response Sequences

1. The user uploads or selects two supported document versions.
2. The system extracts accessible content and prepares the documents for comparison.
3. The comparison component identifies textual differences and supported changes.
4. The system displays the differences in an organized comparison view.
5. The user reviews the changes and may export the comparison results where supported.

4.18.3 Functional Requirements

- **AX-REQ-68:** The system shall compare two supported document versions.
- **AX-REQ-69:** The system shall identify detectable textual additions, deletions, and modifications.
- **AX-REQ-70:** The system should present detected differences in a format that supports user review.
- **AX-REQ-71:** The system shall notify the user when a document format or content element cannot be compared reliably.

Chapter 5

Other Nonfunctional Requirements

5.1 Performance Requirements

The system shall provide responsive interaction under normal operating conditions. Response time and processing duration may vary depending on file size, model response time, OCR complexity, data volume, and workflow length. The system shall display the current status of long-running tasks, including processing, completion, and failure where applicable. It shall handle timeouts and component failures gracefully and shall not leave workflows in an unknown or misleading state. Performance targets for response time, file processing, and concurrent users shall be defined and evaluated during implementation and testing.

5.2 Safety Requirements

AI-generated documents, summaries, calculations, OCR results, formulas, code, and presentations shall be presented for user review before important external use or consequential execution, where applicable. The system shall identify failed, incomplete, or interrupted workflows rather than reporting them as successfully completed. Where supported, the system should provide relevant source information, assumptions, or explanations for analytical results. The system shall communicate uncertainty or processing limitations when they are detected. Automated checks and AI-generated explanations shall not be treated as guarantees of factual correctness. Actions requiring explicit user approval shall not proceed until the required approval has been received.

5.3 Security Requirements

The deployed system shall use HTTPS to protect data transmitted between the client and server. Where user accounts are implemented, authentication and authorization shall protect private files, preferences, and workflow history from unauthorized access. API keys, database credentials, and other secrets shall be stored securely on the server and shall not be exposed in frontend code or public repositories. Uploaded files and generated outputs shall be accessible only to authorized users. The system shall validate uploaded files and apply appropriate restrictions on supported file types and sizes. Sensitive information shall be handled according to the implemented privacy and data-protection mechanisms. The project team shall define file-retention and deletion policies and disclose relevant external AI services that may process user data.

5.4 Software Quality Attributes

- **Usability:** The interface should provide clear navigation and understandable workflows so that users can perform common tasks with minimal training.
- **Reliability:** The system shall report errors and incomplete operations clearly and shall not mark failed operations as successful.

- **Maintainability:** The system should use modular components and well-defined interfaces to facilitate testing, debugging, and future updates.
- **Scalability:** The architecture should support the future addition of AI agents, office operations, and workflow capabilities without requiring a complete system redesign.
- **Interoperability:** The system shall process supported office file formats through compatible libraries, APIs, or defined interfaces.
- **Transparency:** The system should display workflow progress, relevant processing steps, and explanations of AI-generated results where supported.
- **Accessibility:** The system shall provide text-based input for supported tasks so that users are not required to use voice commands when voice interaction is available.
- **Portability:** The application should be deployable in environments compatible with its selected programming languages, libraries, database, and external service dependencies.
- **Privacy:** The system shall apply appropriate access controls and configured data-protection measures when handling user documents and sensitive information.
- **Accuracy:** The system should validate computational results where feasible and inform users of limitations affecting AI-generated content, OCR extraction, and automated analysis.
- **Compatibility:** The web application should function correctly on supported modern browsers, and file-processing features shall be limited to the formats supported by the implemented components.

Chapter 6

Other Requirements

The system shall be designed to support future enhancements, including the integration of additional AI models, office applications, workflow automation capabilities, and external services. It should maintain a modular architecture to simplify maintenance, testing, and upgrades. The system shall provide appropriate error messages and guidance when a requested operation cannot be completed due to unsupported file formats, unavailable services, invalid inputs, or processing failures. User documentation and basic usage instructions should be provided to help users understand the system's features, limitations, and workflow.

The system shall maintain consistency between user requests, workflow execution, and generated outputs. It should provide suitable feedback on task progress and completion and allow users to review AI-generated content before using or applying it. The implementation shall comply with applicable software licensing requirements and the terms of any external APIs, AI models, libraries, or services used. Additional requirements related to deployment, data retention, supported file formats, and integration with office applications shall be finalized according to the selected implementation approach and project scope.

Appendix A: Glossary

Agentic AI	An AI-based system that plans and performs multi-step tasks using configured tools.
AI Agent	A software component assigned a task or role in a workflow.
ASR	Automatic Speech Recognition; conversion of spoken audio into text.
LLM	Large Language Model.
Multimodal AI	AI that processes multiple input types, such as text and images.
OCR	Optical Character Recognition; extraction of text from images or scanned documents.
PPTX	Office Open XML presentation file format.
SRS	Software Requirements Specification.
Workflow orchestration	Coordination of task order, dependencies, state, and results.
XAI	Explainable Artificial Intelligence.

Appendix B: Analysis Models

The final report may include:

- **Use Case Diagram:** Interactions between users and AXIS AI.
- **System Architecture Diagram:** Frontend, backend, AI model, agent framework, OCR/speech components, file tools, and database.
- **Activity Diagram:** User command, planning, execution, checking, and output delivery.
- **Data Flow Diagram:** Movement of commands, uploaded files, intermediate results, and generated outputs.

Appendix C: To Be Determined List

- Final frontend and backend frameworks.
- Final LLM and agent-orchestration framework.
- Final speech-recognition and OCR engines.
- Supported file formats and maximum upload sizes.
- Whether multilingual and handwritten-text support will be included in the first implementation.
- Whether workflow history and saved preferences will be persisted.
- Performance targets and test datasets.
- Data-retention policy and use of external AI services.
- Which proposed features will be implemented in the first prototype.